airiti ARTICLES IN PRESS

指數股票型基金與股市波動：臺灣股票型 ETF 實證分析及申購買回機制轉變和市場動盪之研究

Exchange-Traded Funds and Stock Market Volatility: Analyzing Taiwan Equity ETFs, Creation-Redemption Changes, and Market Turbulence

吳蓩清* E-Ching Wu

靜宜大學財務金融學系

Department of Finance,

Providence University

本文引用格式建議：吳蓩清(2025)。指數股票型基金與股市波動: 臺灣股票型 ETF 實證分析及申購買回機制轉變和市場動盪之研究。中山管理評論，預刊登文章。DOI：10.6160/SYSMR.202503/PP.0002

Suggested Citation: Wu, E. C. (2025). Exchange-Traded Funds and Stock Market Volatility: Analyzing Taiwan Equity ETFs, Creation-Redemption Changes, and Market Turbulence. **Sun Yat-sen Management Review**, Advance online publication. DOI：10.6160/SYSMR.202503/PP.0002

---

* Corresponding Author: E-Ching Wu is an Assistant Professor in the Department of Finance at Providence University, Taiwan. Address: 200, Sec. 7, Taiwan Boulevard, Shale District, Taichung City 43301, Taiwan, R.O.C. Phone: +886-4-26328001 ext. 13606. Email: wuec@pu.edu.tw.

I appreciate the helpful comments from anonymous referees and the editor.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

摘要

本研究以 GARCH、GJR-GARCH 與 E-GARCH 模型，探討指數股票型基金 (ETF) 對臺灣股票市場波動性的影響。研究結果顯示，ETF 流動性增加會使股市波動性上升，意味 ETF 交易活動的增長會吸引噪音交易者投入，進而提升市場不確定性。同時，股票市場本身流動性的提高亦加劇市場波動。此外，實證結果也指出市場對負面訊息的衝擊反應較為強烈，呈現負向不對稱性。

本文亦發現現金申購/買回機制的導入以及在市場動盪時期，均會擴大 ETF 對市場波動的影響。是以，ETF 流動性不僅加劇股票報酬的波動，亦影響風險溢酬的變動，投資人應密切關注 ETF 交易活動的變化，而管理當局也需在市場動盪時期採取適當措施，以穩定金融市場。

關鍵詞：指數股票型基金、波動性、GARCH、周轉率

ABSTRACT

This study explores the impact of exchange-traded funds (ETFs) on stock market volatility, employing GARCH, GJR-GARCH, and E-GARCH models to analyze stock returns on the Taiwan Stock Exchange. The findings show that elevated ETF turnover heightens market volatility, fueled by increased trading activity and noise trader involvement. Greater market liquidity further amplifies this volatility. Moreover, the analysis uncovers a negative asymmetry in market response, with stock returns exhibiting a stronger reaction to negative information shocks than to positive ones.

Furthermore, both the transition in the creation and redemption mechanism and periods of market turbulence intensify the influence of ETFs on stock market volatility. These findings stress the importance for investors to closely monitor ETF trading activity, given its substantial effect on stock return volatility and risk premiums. Regulators should consider adopting appropriate policies to ensure market stability as ETFs play an increasingly prominent role in financial markets.

Keywords: ETF, volatility, GARCH, turnover

JEL Classification: G11, G12

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

1. INTRODUCTION

After Fama (1970) proposed the hypothesis of capital market efficiency, many researchers tested the validity of his proposition, and small investors adopted indexed strategies, assuming that the efficient market hypothesis (EMH) held true (see Brown, 2020). Since the launch of the first exchange-traded fund (ETF), Standard & Poor's Depositary Receipts (SPDR), by the American Stock Exchange in 1993, ETFs have rapidly gained popularity as a result of their accessibility, diversification, and lower transaction costs and taxes burdens.[1] Over the past two decades, the trading volume of ETFs has surged, and their share of the overall stock market value has steadily increased.[2] However, the impact of ETFs on stock market efficiency remains uncertain, primarily owing to the increasing number of free riders, which intensifies market opacity. Qin and Singal (2015) suggest that the expansion of indexing is linked to a decline in stock price efficiency. This effect occurs because index fund trading is driven mainly by investor flows or changes in index composition, rather than the incorporation of private information, thereby adversely affecting price efficiency.

As Ben-David et al. (2017) stressed, ETF prices align with their underlying basket through two key mechanisms: primary and secondary market arbitrage. In the primary market, authorized participants (APs) correct price discrepancies by exchanging lower-priced assets—either ETF shares or the underlying basket—for higher-priced ones, facilitating the creation or redemption of ETF shares. In the secondary market, traders exploit price gaps through long and short positions, ensuring alignment between ETF prices and the value of the underlying securities.

This profit-driven process yields contrasting effects. On the one hand, it enhances liquidity in the underlying assets, improving price discovery and advancing their informational efficiency (e.g., Guzman et al., 2022). On the other hand, it can attract speculative "noise trading," which may introduce inefficiencies (e.g., Broman, 2022). These dual and opposing effects have made the impact of ETFs on stock market efficiency a subject of considerable academic interest, sparking ongoing research into their broader market implications.

[1] For example, the trading tax rate for ETFs in the Taiwan stock market is 1‰, while that for common stocks is 3‰.

[2] Elton et al. (2019) noted that the proportion of assets under management in passive funds increased from 10.5% to 26% between 2007 and 2016. Similarly, in Taiwan, the market value of ETFs expanded from 89,414 million NTD in 2014 to 1,411,133 million NTD in 2023. Consequently, the share of ETF market value relative to the total stock market value increased from 0.375% in 2014 to 2.495% in 2023.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

The existing literature has extensively explored the impact of ETFs on the liquidity of their constituent stocks. Karmaziene and Sokolovski (2022) demonstrated that ETFs alleviate short-sale constraints on the liquidity of component stocks. Likewise, Li and Zhao (2014) found that the introduction of leveraged ETFs increases the liquidity of constituent stocks. Beyond liquidity effects, the influence of ETFs on stock market volatility has attracted considerable scholarly attention. ETFs are widely regarded as instruments that enhance financial market efficiency by enabling investors to respond more swiftly to new information than they could when trading the underlying securities directly (e.g., Lettau & Madhavan, 2018; Madhavan, 2016). Glosten et al. (2021) argued that ETF activity strengthens informational efficiency by improving the alignment between short-term fundamentals and stock prices. This enhanced informational efficiency supports price discovery and may help mitigate market volatility. Furthermore, a theoretical framework proposed by Chabakauri and Rytchkov (2021) suggests that as some investors transition from trading individual stocks to index-based trading, market return volatility declines.

However, empirical evidence directly linking ETFs to reduced stock market volatility remains limited. For instance, Li and Zhao (2014) found that the volatility of constituent stocks was unaffected by leveraged ETF trading and, in fact, declined following the introduction of ETFs. While ETFs may reduce idiosyncratic risk for individual component stocks, they can increase systematic risk while leaving their effect on information efficiency and market volatility ambiguous (Kim & Kim, 2023). Marta and Riva (2025) find that the transition from synthetic to physical replication in ETFs reduces the variance ratio of component stocks.

While extensive literature suggests that ETFs reduce the information efficiency of their component securities, studies such as Broman (2016) and Brown et al. (2021) indicate that ETF investors often underperform because of non-fundamental demand and poor market timing. Ben-David et al. (2018) observed that ETFs exhibit high share turnover, with trading primarily driven by short-term investors who frequently engage in directional trading and hold securities only briefly. This behavior positions ETFs as cost-effective instruments for executing short-term speculative strategies. Consequently, these dynamics suggest that ETFs may amplify stock market volatility. Empirical evidence from multiple studies (e.g., Anadu et al., 2020; Ben-David et al., 2018; Dedi et al., 2016; Rossi & Steliaros, 2022) supports this assertion, indicating that ETFs contribute to increased market volatility.

The empirical evidence on the impact of ETFs on stock market volatility remains inconclusive, presumably caused by variations in product types, research

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

methodologies, and the specific subjects of analysis. For instance, Li and Zhao (2014) found that leveraged ETF trading did not affect the return volatility of basket stocks[3], whereas Rossi and Steliaros (2022) observed that the inclusion of ETFs in stocks heightened volatility, particularly during the market closing auction phase. While most existing studies have primarily focused on the volatility of ETFs' constituent stocks, this study shifts the analytical focus to the underlying indices of ETFs to examine whether their presence influences overall stock market volatility. The findings indicate that ETF turnover has a positive impact on stock market volatility, corresponding to previous research while providing deeper insights into the evolving role of ETFs in financial markets.

ETFs were first introduced to the Taiwanese stock market on June 30, 2003. Over the past three years, their total market value in Taiwan has surged significantly, reaching NTD 5.29 trillion as of June 2024. ETFs now account for 62.95% of the mutual fund market, positioning Taiwan as the third-largest ETF market in Asia by value.[4] Taiwan's remarkable growth and extensive history of ETF development have been shaped by several pivotal events, including transitions in ETF creation and redemption mechanisms and market disruptions caused by the COVID-19 pandemic. This continuous evolution presents a crucial opportunity to reassess the validity of prior research findings within the context of changing market conditions.

The creation and redemption mechanisms of ETFs are broadly categorized into physical and cash-based processes. Cash mechanisms, typically employed by synthetic ETFs, are associated with counterparty default risk in derivative transactions (Ben-David et al., 2017). Compared to physical mechanisms, cash mechanisms facilitate easier participation and tend to attract more arbitrageurs. This increased arbitrage activity can lead to noisier trades and heightened market volatility. As Marta and Riva (2025) found, the transition from synthetic to physical replication in ETFs reduces the variance ratio of component stocks. This suggests that shifting the creation and redemption mechanism from physical to cash increases stock volatility. In Taiwan's ETF market, the cash creation and redemption mechanism were introduced on November 1, 2016, complementing the existing physical mechanism. This development

[3] Li and Zhao (2014) employed an event study methodology to examine the impact of leveraged ETFs on the volatility of their component stocks, concluding that the volatility remained unaffected by the presence of these ETFs. This finding may stem from differences in the methodologies and subjects observed, as their approach contrasts with other studies that utilize GARCH models and focus on various ETF products.

[4] Refer to the press release dated July 30, 2024, on the homepage of the Financial Supervisory Commission in Taiwan.
https://www.fsc.gov.tw/ch/home.jsp?id=2&parentpath=0&mcustomize=news_view.jsp&dataserno=202407300002&dtable=News

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

presents a unique opportunity to examine the impact of this dual mechanism structure on stock market volatility, representing a key contribution of this study. The findings illustrate that the adoption of the cash mechanism significantly amplifies the influence of ETFs on market volatility, offering profound understanding of their role in financial market dynamics.

Moreover, during periods of market turbulence, the scarcity of reliable information often prompts market makers and arbitrageurs to suspend their intermediation activities. This withdrawal aggravates illiquidity in underlying securities, amplifying market shocks and propagating volatility to other asset classes (Ben-David et al., 2017). Several studies have documented this amplification effect, wherein arbitrageurs or hedge fund investors exiting the market contribute to increased stock market volatility (e.g., Aragon & Strahan, 2012; Ben-David et al., 2012). Given that the sample period in this study encompasses the COVID-19 pandemic, it offers a valuable opportunity to re-examine the amplification effect of market turbulence on ETFs and their influence on volatility. The findings reaffirm illuminating revelations while offering deeper perspectives that further enrich the academic discourse on this topic.

Prior research on stock volatility and ETFs has predominantly utilized the GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model to explore their relationship. Furthermore, studies have emphasized that the GJR-GARCH (Glosten-Jagannathan-Runkle GARCH) and E-GARCH (Exponential GARCH) models are particularly well-suited for capturing the asymmetries commonly observed in stock return volatility (Dedi et al., 2016; Duan & Lin, 2014). Building on this foundation, this study employs three GARCH variants—GARCH, GJR-GARCH, and E-GARCH—to investigate the influence of ETFs on stock market volatility. This comprehensive approach strengthens the robustness of the empirical findings while offering valuable insights into the information efficiency of the market.

In summary, this study provides greater clarity on the positive relationship between ETF turnover and market volatility, demonstrating that the influence of ETFs on stock market fluctuations is further expanded by both transitions in creation and redemption mechanisms and periods of market turbulence. Moreover, this research contributes to the broader literature by offering penetrating observations into the intricate relationship between ETF turnover ratios and market volatility, enhancing our understanding of the evolving dynamics of financial markets.

The remainder of this paper is organized as follows. Section 2 presents the development of hypotheses, which serve as the foundation for the subsequent empirical

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

analysis. Section 3 details the methodology, including the empirical models, the expected effects of relevant variables, and the data collection process. Section 4 provides a comprehensive discussion of the empirical results. Finally, Section 5 concludes the analysis by summarizing key findings and offering in-depth focus on and recommendations for investment strategies and regulatory policies.

2. HYPOTHESIS DEVELOPMENT

As discussed in Section 1, a substantial body of literature suggests that the introduction or trading activity of ETFs contributes to increased stock market volatility. This conclusion is grounded in the mechanism by which demand for ETF shares triggers rapid price adjustments in the underlying component stocks, thereby amplifying volatility (Ben-David et al., 2018). Stock market volatility is further influenced by the creation and redemption processes of ETFs. When deviations occur between an ETF's market price and its net asset value (NAV), investors seek to capitalize on these discrepancies, intensifying market dynamics.

Moreover, the inherent features of ETFs, such as seamless and continuous market access combined with low transaction costs, make them attractive to speculators and arbitrageurs, leading to high-frequency trading activity. Consequently, the trading frequency of ETFs, often quantified by their turnover ratio, is positively correlated with heightened stock market volatility. The turnover ratio also serves as a proxy for trading volume and liquidity, as noted by Avgouleas & Degiannakis (2009), Chordia et al. (2001), Datar et al. (1998), and Tkac (1999)[5].

In light of this reasoning, the following hypothesis is proposed to explore the relationship between ETF trading activity and stock market volatility.

Hypothesis 1 (H1)*: The turnover of ETFs positively influences stock return volatility.*

Following the same reasoning, the trading frequency of the stock market, as represented by its turnover ratio, is anticipated to affect stock market volatility in a comparable manner. Based on this expectation, the following hypothesis is proposed to examine the impact of stock market turnover on volatility.

[5] Avgouleas & Degiannakis (2009) identified three key measures of trading volume: volume-based turnover, value-based turnover, and the ratio of value-based turnover to market value. The latter, equivalent to the ratio of the number of shares traded to the total number of shares outstanding, aligns with the definition adopted in this study. This measure overcomes the limitations of the other two, which are impacted by market fluctuations that cause changes in stock prices and the number of outstanding shares. Datar et al. (1998) also utilized this turnover ratio as a proxy for liquidity, supporting its relevance in examining market dynamics.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Hypothesis 2 (H2)*: Stock market turnover positively impacts stock return volatility.*

Building on the previously stated hypotheses, the following section outlines the methodology utilized in the subsequent empirical analysis.

3. METHODOLOGY

This study is intended to examine the impact of ETF growth on stock volatility. This section introduces three GARCH models used to estimate stock return volatility and to observe the influence of ETFs. The discussion includes definitions of the variables relevant to each model. Furthermore, the three GARCH models and their corresponding variance equations, employed in the empirical analysis, are presented. Finally, the anticipated effects of the relevant variables, as predicted by the hypotheses, are outlined.

3.1 Variance Equation Models

Since Engle (1982) introduced the ARCH model to explain volatility clustering and improve volatility forecasting, Bollerslev (1986) extended it to the GARCH model by incorporating lagged conditional variances and squared residuals. Subsequently, Bollerslev (1987) demonstrated that the GARCH(1,1) model effectively captures the dynamics of stock market volatility while raising the question of whether alternative conditional error distributions might better describe these processes.

Engle & Ng (1993) compared PNP-ARCH, GJR-GARCH, and E-GARCH models for Japanese stock returns, finding GJR-GARCH best captured negative shocks and E-GARCH effectively modeled asymmetries. Moreover, numerous studies have further established the superiority of asymmetric GARCH models over symmetric ones. For instance, research has demonstrated the fitting advantages of GJR-GARCH (e.g., Alberg et al., 2008; Duan & Lin, 2014) and E-GARCH (e.g., Dedi et al., 2016; Nelson, 1991) models.

Building upon this foundation, the present study employs the GARCH model to examine the influence of ETFs on market volatility, serving as a benchmark for further analysis. Additionally, the GJR-GARCH and E-GARCH models are utilized to investigate the robustness of the impact of ETFs on market volatility.

3.1.1 GARCH model

Building on extensive prior research, this study models the volatility of stock returns using a GARCH(1,1) process. To evaluate the impact of exchange-traded funds (ETFs), the ETF turnover ratio is incorporated into the variance equation. Recognizing

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

the established influence of stock market liquidity on volatility, as highlighted by studies such as Kapusuzoglu and Ceylan (2018) and Lamoureux and Lastrapes (1990), the turnover ratio of the stock index is also included. Furthermore, Duan and Lin (2014) emphasize that implied volatility provides reliable information about the returns of underlying assets in options and captures substantially more information about future market volatility. This insight supports the inclusion of the squared VIX index from the preceding period in the variance equation. Accordingly, this study integrates these key variables—ETF turnover, stock market turnover, and the squared previous-period VIX—into the variance equation to provide a comprehensive analysis. The GARCH model is demonstrated as follows:

$$R_t = \alpha_0 + \alpha_1 R_{t-1} + \varepsilon_t \ , \quad \varepsilon_t | \phi_{t-1} \sim N(0, \sigma_t^2) \tag{1}$$

$$\sigma_t^2 = \theta_0 + \theta_1 \varepsilon_{t-1}^2 + \omega_1 \sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 \tag{2}$$

where R_t represents the return of the stock index, σ_t^2 denotes the conditional variance, $TR_{ETF,t}$ is the turnover ratio of the ETFs, $TR_{mkt,t}$ represents the turnover ratio of the stock index, and VX_{t-1}^2 denotes the square of the VIX index.

Equation (1) represents the mean equation, describing the relationship between the stock index return and its previous return. Equation (2) defines the variance equation, incorporating stock return volatility and related variables such as the turnover ratio of ETFs, the turnover ratio of the stock index, and the square of the VIX index from the previous period.

3.1.2 GJR-GARCH model

As delineated in the preceding discussion, the GJR-GARCH model demonstrates superior capability compared to the standard GARCH model in capturing the asymmetric responses of stock returns to market shocks. Subsequently, this study adopts the GJR-GARCH model to more accurately characterize stock return volatility and to re-evaluate the impact of ETFs on stock market dynamics. The representative variance equation in the GJR-GARCH (1,1,1) model is presented as follows:

$$\sigma_t^2 = \theta_0 + \theta_1 \varepsilon_{t-1}^2 + \gamma_1 \varepsilon_{t-1}^2 DV_{t-1}^- + \omega_1 \sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$$

$$DV_{t-1}^- = \begin{cases} 1 \ , if \ \ \varepsilon_{t-1} < 0 \\ 0 \ , otherwise \end{cases} \tag{3}$$

where DV_{t-1}^- is a dummy variable that captures the response of stock volatility to asymmetric shocks in stock returns, where negative forecast errors in stock returns generate greater volatility. This indicates that the stock market is more responsive to

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

"bad" news than to "good" news.

3.1.3 E-GARCH model

Similarly, research comparing the fit of different GARCH models to stock volatility has shown that the E-GARCH model is also well-suited for capturing the asymmetric characteristics of stock return shocks. Thus, this research also employs the E-GARCH (1,1) model to evaluate the impact of ETF turnover on stock return volatility. The variance equation used for estimating stock volatility is presented below.

$$\ln(\sigma_t^2) = \theta_0 + \varphi_1 \left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \varphi_2 \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \varphi_3 \ln(\sigma_{t-1}^2) + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 \quad (4)$$

3.2 Modified Mean Equation

Previous studies (e.g., Duan & Lin, 2014; Ibbotson et al., 2013; Kapusuzoglu & Ceylan, 2018) have demonstrated that stock returns are significantly influenced by stock market liquidity, often measured through turnover or trading volume. In due course, this analysis integrates the turnover of the overall stock market—calculated as trading volume relative to total outstanding shares—into the mean equation, as specified in Equation (5). The system of equations is then re-estimated to determine whether the inclusion of this variable alters the observed impact of ETFs. The revised mean equation is presented below.

$$R_t = \alpha_0 + \alpha_1 R_{t-1} + \alpha_2 TR_{mkt,t} + \varepsilon_t \ , \quad \varepsilon_t|\phi_{t-1} \sim N(0, \sigma_t^2) \quad (5)$$

Following Equation (5), the system of equations is re-estimated to perform a robustness check.

Based on Hypotheses 1 and 2, the coefficients of the independent variables representing ETF turnover and stock market turnover in the variance equations of the GARCH, GJR-GARCH, and E-GARCH models are anticipated to meet the following conditions:

$$\beta_1 > 0, \quad \beta_2 > 0$$

Additionally, if the stock return exhibits a negative asymmetry in response to market shocks, the coefficients of the associated independent variables DV_{t-1}^-, $\frac{\varepsilon_{t-1}}{\sigma_{t-1}}$ are expected to be positive and negative, respectively, implying $\gamma_1 > 0$, $\varphi_2 < 0$.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

This analysis employs the previously outlined empirical models and hypotheses, utilizing data sourced from the Taiwan Economic Journal (TEJ) and conducting the empirical evaluation using EViews 10 econometric software.

3.3 Data

The data for the empirical analysis is obtained from the Taiwan Economic Journal (TEJ) covering the period from January 1, 2014, to December 31, 2023, comprising 2,445 business days. Due to the use of lagged data in the system estimation, the effective sample size for each series consists of 2,444 observations. The variables included in the model were processed as follows to prepare for regression analysis:

1. R_t is calculated from the stock index as the natural logarithm of $(\frac{P_t}{P_{t-1}})$, where P_t represents the stock index at time *t*.

2. $TR_{EFT,t}$ is calculated as the natural logarithm of the daily trading volume of ETFs divided by the total number of outstanding shares, following the definition provided by Chordia et al. (2001).

The first ETF on the Taiwan Stock Exchange (TWSE) was introduced on June 30, 2003. Despite this, the growth of ETFs tied to domestic equity has remained limited. As of July 17, 2017, only 10 domestic equity ETFs were listed on the TWSE. For this empirical analysis, the ETFs were selected based on their market value ratios—the fund's market value relative to the total stock market value—over the 10-year sample period (2014–2023). These average ratios ranged from 0.0007% to 0.3079% (see Appendix Table A1). Among them, only the Yuanta/P-shares Taiwan Top 50 ETF (0050) and the Yuanta/P-shares Taiwan Dividend Plus ETF (0056) recorded average market value ratios exceeding 0.10%. In addition, both ETFs had trading value ratios above 0.16%, reinforcing their selection.

Due to the significant market share of these two ETFs, compared to other domestic equity ETFs, ETF turnover is constructed using these two funds. Tkac (1999) defines the associated market turnover as the total shares traded divided by the total shares outstanding, representing a share-weighted average of individual firm turnover ratios.

To address potential issues of sample overlap, the turnover of 0050 and 0056 was calculated separately using their respective trading volumes and outstanding shares. This ensures that the analysis accurately captures each ETF's individual contribution.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

3. $TR_{mkt,t}$ represents the turnover of the entire stock market at time t, derived directly from the TEJ databank. It is calculated as the natural logarithm of the trading volume divided by the number of shares outstanding.

4. VX_{t-1} is the TAIEX Options Volatility Index at the close of trading on day t-1, obtained from the TEJ databank and the Taiwan Futures Exchange (TAIFEX).

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

4. EMPIRICAL RESULTS

To conduct the empirical analysis, this research follows a systematic procedure for data collection and subsequently regresses the data on the specified empirical models. The empirical results are presented in this section.

4.1 Tests for Normality and Stationarity

Prior to performing the regression analysis, it is crucial to ensure that the probability distribution of R_t is non-normal and that the R_t process is stationary. To achieve this, descriptive statistics are presented in Table 1, and the stock index return series is illustrated in Figures 1 and 2, offering a comprehensive overview of the characteristics of stock index returns. Following this preliminary assessment, the normality of the data is tested using the Jarque-Bera (JB) statistic, while stationarity is evaluated through the ADF, PP, and KPSS unit root tests. To validate the suitability of the GARCH, GJR-GARCH, and E-GARCH models, serial correlation and hetero-skedasticity are examined using the Ljung-Box Q and Ljung-Box Q^2 statistics, respectively. The outcomes of these tests are detailed in Tables 2 to 4.

Table 1. Descriptive Statistics for stock return, turnover of ETFs and stock market

Unit: %

	Stock index return	Turnover of ETFs	Turnover of market	Turnover of 0050	Turnover of 0056
Mean	0.0300	0.8525	0.4127	0.8575	0.7944
Median	0.0716	0.6599	0.3516	0.6594	0.5344
Maximum	6.1726	11.4460	2.0876	10.8811	13.5423
Minimum	-6.5206	0.1157	0.0969	0.0950	0.0572
Std. Dev.	0.9523	0.7082	0.2048	0.7269	0.8813
CV	31.7490	0.8307	0.4963	0.8477	1.1094
Observations	2445	2445	2445	2445	2445

As shown in Table 1, the mean stock return is 0.03% with a standard deviation of 0.9523%, resulting in a coefficient of variation (CV) of 31.749 (0.9523/0.03). This

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

notably high CV reflects substantial variability in stock returns, potentially indicating the presence of heteroskedasticity.

Regarding turnover ratios, the average turnover ratio of ETFs (0.8525%) surpassed that of the stock market (0.4127%). Moreover, ETFs display greater turnover variability, as indicated by a higher CV (0.831) than stocks (CV = 0.496), underscoring their relative volatility associated with ETFs. The turnover pattern of ETFs 0050 mirrors that of ETFs in general but is more volatile, with a CV of 0.8477—higher than the overall ETF CV. In contrast, ETF 0056 has a lower average turnover than both of 0050 and the overall ETF, yet it exhibits the highest volatility among the three ETFs, as measured by its CV.

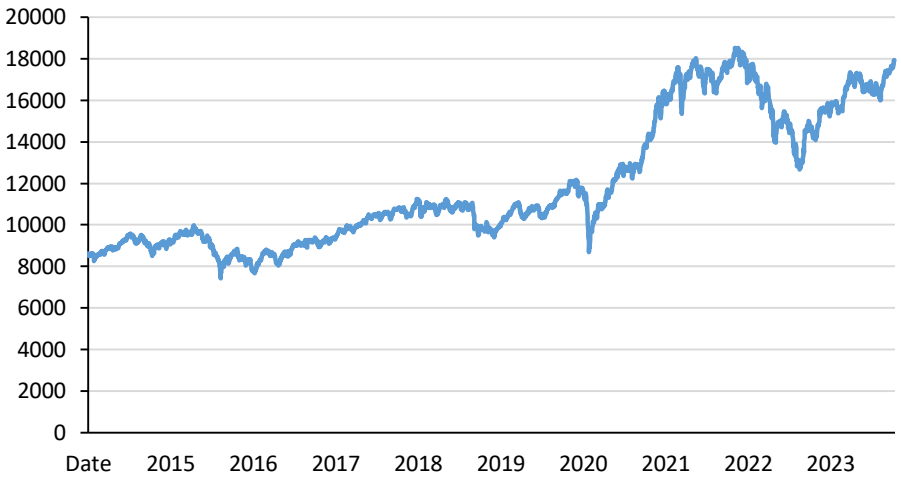


Figure 1. Daily stock index in TWSE (TAIEX)

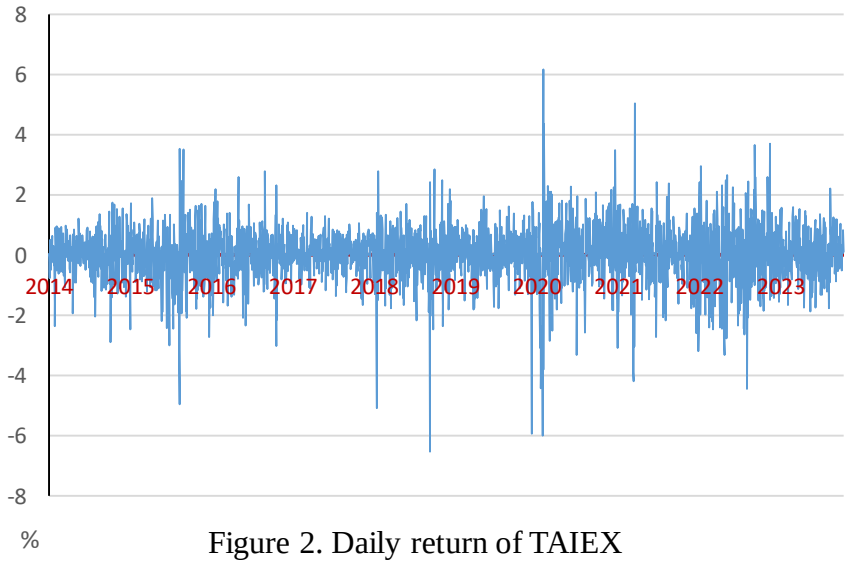


Figure 2. Daily return of TAIEX

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Figures 1 and 2 clearly demonstrate that the return process of the stock index in TWSE exhibits stationarity and volatility clustering. The observed volatility in stock returns is appropriately modeled using GARCH models. To validate these observations, the normality of the stock return process was tested using the Jarque-Bera (JB) statistic, stationarity was examined using the ADF, PP, and KPSS unit root tests, and serial correlation and heteroskedasticity were assessed through the Ljung-Box Q and Ljung-Box Q^2 statistics. The results of these analyses are presented in the following tables.

Table 2. The results of normality test on stock index return process

	Return of stock index
Skewness	-0.5689
Kurtosis	8.2087
Jarque-Bera	2895.764***
	(0.0000)

Note: ( ) is the p-value. ***means significance at 1% level.

The stock index return process does not follow a normal distribution, as indicated by its negative skewness (-0.5689) and fat-tailed distribution (kurtosis = 8.2087), as shown in Table 2. Besides, the Jarque-Bera statistic of 2895.764, with a p-value significant at the 1% level, rejects the null hypothesis of normality. Consequently, the process is not suitable for regression using the OLS method.

Table 3. The results of stationary tests on stock index return process

Unit root test		ADF	PP	KPSS
t-Statistics/LM for KPSS		-32.7879***	-48.8681***	0.0685
Test critical values:	1% level	-3.4328	-3.4328	0.739
	5% level	-2.8625	-2.8625	0.463
	10% level	-2.5673	-2.5673	0.347

Note: ***means significance at 1% level and rejects the null hypothesis: the return of stock index has a unit root. While the null hypothesis of KPSS: the return of stock index is stationary.

As shown in Table 3, the t-statistics for the ADF and PP tests are -32.79 and -48.87, respectively. Both reject the null hypothesis, indicating that the process of stock index return is stationary. More than that, the KPSS test yields an LM statistic of 0.0685,

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

which fails to reject the null hypothesis of the stock return being stationary. The stock index return process is thus confirmed to be a stationary series.

Before proceeding with the regression of GARCH models, it is necessary to diagnose the serial correlation and heteroskedasticity in the stock index returns. These are assessed using the Ljung-Box Q and Q^2 statistics. Given that two mean equations are used in the estimation, the Ljung-Box tests are conducted twice. The results of these tests are presented in Table 4.

Table 4. Tests of the serial correlation and heteroskedasticity of index return residuals

	mean equation (1)		mean equation (5)	
	serial correlation test			
lag periods	Q-Statistics	Probability	Q-Statistics	Probability
10	17.237*	0.069	18.452**	0.048
20	25.035	0.2	25.578	0.18
	heteroskedasticity test			
lag periods	Q^2-Statistics	Probability	Q^2-Statistics	Probability
10	748.47 ***	0	738.65 ***	0
20	774.71 ***	0	767.26 ***	0

Note: ***, ** and * indicate significance at 1%, 5% and 10% separately. Equation (1): $R_t = \alpha_0 + \alpha_1 R_{t-1} + \varepsilon_t$, equation (5): $R_t = \alpha_0 + \alpha_1 R_{t-1} + \alpha_2 TR_{mkt,t} + \varepsilon_t$.

As shown in Table 4, the Q statistics of equations (1) and (5) at a lag of 10 periods are 17.237 and 18.452, respectively, and are statistically significant. However, at a lag of 20 periods, these statistics are insignificant. This indicates that the residuals of the stock index return exhibit serial correlation. What's more, the Q^2 statistics for equation (1) at lags of 10 and 20 periods are 748.47 and 774.71, respectively, while for equation (5) they are 738.65 and 767.26, all of which are significant at the 1% level. This confirms the presence of heteroskedasticity in the residual series of stock returns.

Given that the stock index return series exhibits both serial correlation and heteroskedasticity, it is appropriate to proceed with estimating the system of equations using the GARCH, GJR-GARCH, and E-GARCH models.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

4.2 Empirical Results

Since the stock index return process met the prerequisites for GARCH model regression, the empirical analysis was conducted using EViews 10, with the results summarized in Tables 5 and 6.

4.2.1 Case of autocorrelated mean equation

As expounded in Section 3.1, the stock index return (R_t) was regressed on R_{t-1} using the GARCH, GJR-GARCH, and E-GARCH models. The variance equation was structured with four different cases: (1) including only $TR_{ETF,t}$, (2) including both $TR_{ETF,t}$ and $TR_{mkt,t}$, (3) including $TR_{ETF,t}$ and VX_{t-1}^2, and (4) including all three regressors ($TR_{ETF,t}$, $TR_{mkt,t}$ and VX_{t-1}^2) sequentially. The variance equation was therefore tested in four distinct forms, with the empirical results summarized in Table 5.

To address the issue of overlapping samples arising from the intersection of ETF turnover for the constituent stocks of 0050 and 0056, the regression analysis was adjusted by substituting the turnover of 0050 and 0056 individually into the ETF turnover. The empirical results from these adjustments were compared with the original findings to evaluate whether the impact of ETFs on stock market volatility remained consistent. The results of this analysis, focusing on the turnover of 0050 and 0056, are presented in Tables A2 and A3 in the Appendix.

Consistent with the hypotheses outlined in Section 2, the coefficients in the variance equations are expected to satisfy the conditions $\beta_1 > 0$, $\beta_2 > 0$, $\gamma_1 > 0$ and $\varphi_2 < 0$. Table 5 presents the empirical results for the coefficients within the system of equations. The findings show that the mean equation coefficients (α_1) are all negative across different variance equation specifications in the GARCH models except in models C4 and C8. In the E-GARCH model, these coefficients are statistically significant at the 1% level.

Examining the coefficients in the variance equations across different GARCH model settings, the coefficients for $TR_{ETF}(\beta_1)$ and $TR_{mkt}(\beta_2)$ are consistently positive and significant at the 1% level, correlating with the expectations hypothesized in Section 2. For instance, the β_1 coefficients in models C1, C5, and C9 are 0.3616, 0.3536, and 1.1839, respectively, while the β_2 coefficients in models C2, C6, and C10 are 0.8137, 0.7958, and 1.2212, respectively. In models C4, C8, and C12, which account for the impact of past volatility, the coefficients for both β_1 and β_2 remain positive. Furthermore, the inclusion of stock market liquidity does not alter the sign or statistical significance of β_1. For example, β_1 in model C1 is 0.3616 and significant at the 1%

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

level, while in model C2 it increases to 0.5254, maintaining its 1% significance. These findings are consistent with Hypothesis 1, indicating that stock market volatility intensifies with higher ETF turnover and increased stock market liquidity.

The coefficients representing negative asymmetry in stock return responses interact with findings from previous literature, with $\gamma_1 > 0$ and $\varphi_2 < 0$. In the GJR-GARCH models (C5, C6, C7, C8), the γ_1 coefficients and their corresponding Z-statistics are 0.2387 (13.93), 0.0878 (3.63), 0.1904 (10.4), and 0.0519 (3.89), respectively, all of which are positive and statistically significant at the 1% level. Similarly, in the E-GARCH models (C9, C10, C11, C12), the φ_2 coefficients are negative and significant at the 1% level, with values of -0.0558, -0.0749, -0.1455, and -0.1507, correspondingly. These findings confirm that stock return responses to market shocks exhibit negative asymmetry.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Table 5. Empirical results of mean equation and GARCH, GJR-GARCH, E-GARCH models

regressors in var. / coefficient.	mean equation: $R_t = \alpha_0 + \alpha_1 R_{t-1} + \varepsilon_t$											
	TR_{ETF}	TR_{ETF}, TR_{mkt}	TR_{ETF},VX	TR_{ETF},VX TR_{mkt}	TR_{ETF}	TR_{ETF}, TR_{mkt}	TR_{ETF},VX	TR_{ETF},VX TR_{mkt}	TR_{ETF}	TR_{ETF}, TR_{mkt}	TR_{ETF},VX	TR_{ETF},VX TR_{mkt}
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
α_0	0.0550***	0.0550	0.0730***	0.0414*	0.0427***	0.0537*	0.0550***	0.0460**	0.0648***	0.0589***	0.0592***	0.0526***
	(3.65)	(1.58)	(5.42)	(1.68)	(2.62)	(1.82)	(3.18)	(2.27)	(4.85)	(4.38)	(4.51)	(3.96)
α_1	-0.0263	-0.0219	-0.0571**	0.0074	-0.0104	-0.0030	-0.0246	0.0009	-0.0617***	-0.0637***	-0.0604***	-0.0691***
	(-1.20)	(-0.92)	(-2.50)	(0.38)	(-0.44)	(-0.11)	(-1.00)	(0.05)	(-3.13)	(-3.20)	(-3.34)	(-3.69)
	GARCH				GJR-GARCH				E-GARCH			
θ_0	0.0461***	0.6266***	0.4241***	0.6583***	0.1282***	0.4817***	0.2995***	0.6446***	-0.1064***	-0.1196***	-1.3173***	-1.2960***
	(13.54)	(4.89)	(26.59)	(4.17)	(94.21)	(3.63)	(7.52)	(5.55)	(-9.54)	(-8.62)	(-8.10)	(-7.51)
θ_1/φ_1	0.0750***	0.0042**	0.2388***	0.0237***	0.0297**	-0.0099	0.0585***	0.0150	0.1243***	0.1381***	0.0831***	0.0817**
	(10.55)	(1.96)	(10.66)	(5.02)	(2.09)	(-0.64)	(8.25)	(1.45)	(8.83)	(8.18)	(2.76)	(2.52)
γ_1/φ_2					0.2387***	0.0878***	0.1904***	0.0519***	-0.0558***	-0.0749***	-0.1455***	-0.1507***
					(13.93)	(3.63)	(10.40)	(3.89)	(-5.27)	(-6.75)	(-7.36)	(-7.63)
ω_1/φ_3	0.8537***	0.5649***	0.6120***	0.5758***	0.6686***	0.5576***	0.6657***	0.5758***	0.9863***	0.9836***	0.7925***	0.7994***
	(134.00)	(6.73)	(25.15)	(6.38)	(54.83)	(4.60)	(25.21)	(8.97)	(251.84)	(207.98)	(33.14)	(31.47)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

TR_{ETF}	0.3616***	0.5254***	0.3902***	0.3045***	0.3536***	0.2682***	0.3862***	0.1917***	1.1839***	0.9846***	1.2504***	1.0047***
	(31.01)	(4.31)	(38.36)	(4.40)	(45.75)	(2.68)	(32.55)	(3.64)	(22.13)	(14.53)	(25.62)	(14.16)
TR_{mkt}		0.8137***		0.6118***		0.7958***		0.5397***		1.2212***		1.3696***
		(4.59)		(5.23)		(4.60)		(5.15)		(6.24)		(6.98)
VX_{t-1}^2			-0.0529***	-0.0461**			-0.0295***	-0.0603***			0.2037***	0.2000***
			(-33.43)	(-2.27)			(-3.68)	(-3.70)			(7.57)	(7.07)

Note: ( ) represents the Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, separately.

GARCH: $\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \omega_1\sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$, GJR-GARCH:$\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \gamma_1\varepsilon_{t-1}^2 DV_{t-1}^- + \omega_1\sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$, E-GARCH: $\ln(\sigma_t^2) = \theta_0 + \varphi_1\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \varphi_2\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \varphi_3\ln(\sigma_{t-1}^2) + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$. Cx : the case of different regressors in the variance equation and GARCH methods.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

The results in Table A2 in the Appendix confirm that the coefficients for ETF turnover for both 0050 and 0056 remain consistently positive and highly significant at the 1% level across all GARCH model specifications, providing robust support for Hypothesis 1. Similarly, the findings for stock market turnover remain consistent with the original analysis, showing a positive and significant impact at the 1% level, supporting Hypothesis 2. Additionally, the coefficients capturing negative asymmetry in response to market shocks, γ_1 and φ_2, meet the expected conditions ($\gamma_1 > 0$, $\varphi_2 < 0$) and are statistically significant at the 1% level. Overall, these results reaffirm the consistent influence of ETF turnover and stock market turnover on stock market volatility and they likewise validate the existence of negative asymmetry in responses to market shocks.

4.2.2 Case of the turnover in stock market adding into mean equation

Building on prior research, the return of the stock index is also influenced by its liquidity, represented by the turnover ratio of the stock market index (TR_{mkt}). To incorporate this factor, the mean equation is adjusted to include TR_{mkt}, yielding the following model:

$$R_t = \alpha_0 + \alpha_1 R_{t-1} + \alpha_2 TR_{mkt,t} + \varepsilon_t.$$

In this analysis, the modified mean equation is applied, and the system of equations is re-estimated to assess the robustness of the previous findings. This robustness check is supported by the updated empirical results, which are presented in Table 6.

As discussed in Subsection 4.2.1, this paper examines the effect of ETFs on stock market return volatility, with the coefficients of the turnover ratios for ETFs and the stock market (β_1, β_2) serving as the key variables of interest. Table 6 shows that β_1 is consistently positive and significant at the 1% level across all conditions. For example, the β_1 values for C1, C5, C9, and C4, C8, C12 are 0.3631, 0.3568, 1.1905, and 0.3338, 0.3710, 1.0051, respectively, with corresponding Z-statistics of 30.37, 36.06, 22.2, and 3.73, 19.11, 14.26. Similarly, β_2 is positive and statistically significant at the 1% level in all cases, except in C8, where it is 5% significant.

Furthermore, the β_1 coefficients maintain their positive significance even when accounting for the effect of stock market liquidity. For example, β_1 in C1 is 0.3631 and significant at the 1% level, while in C2, it is 0.3395 and also 1% significant. These results are consistent with the findings from the previous subsection and echo Hypotheses 1 and 2, reaffirming that the growth of ETFs and increased stock market liquidity contribute to higher stock market volatility.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Table 6. Empirical results of mean equation including TR_{mkt} and GARCH, GJR-GARCH, E-GARCH models

regressors in var. / coefficient.	mean equation: $R_t = \alpha_0 + \alpha_1 R_{t-1} + \alpha_2 TR_{mkt,t} + \varepsilon_t$											
	TR_{ETF}	TR_{ETF}, TR_{mkt}	TR_{ETF},VX	TR_{ETF},VX TR_{mkt}	TR_{ETF}	TR_{ETF}, TR_{mkt}	TR_{ETF},VX	TR_{ETF},VX TR_{mkt}	TR_{ETF}	TR_{ETF}, TR_{mkt}	TR_{ETF},VX	TR_{ETF},VX TR_{mkt}
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
α_0	0.0538***	0.0993***	0.0629***	0.0817***	0.0266*	0.0251	0.0466***	0.0339*	0.0682***	0.0644***	0.0619***	0.0569***
	(3.56)	(5.71)	(3.38)	(3.00)	(1.91)	(0.89)	(2.75)	(1.88)	(5.11)	(4.76)	(4.71)	(4.17)
α_1	-0.0250	-0.0220	-0.0368***	-0.0328	-0.0101	-0.0230	-0.0222	-0.0248	-0.0647***	-0.0671***	-0.0610***	-0.0697***
	(-1.14)	(-0.91)	(-2.95)	(-1.23)	(-0.43)	(-0.92)	(-0.93)	(-1.04)	(-3.28)	(-3.36)	(-3.37)	(-3.72)
α_2	-0.1074	0.0547	-0.2171***	0.0244	-0.0290	-0.0415	-0.0103	-0.0267	0.1427*	0.1394*	0.1173	0.0890
	(-1.33)	(0.67)	(-2.86)	(0.20)	(-0.38)	(-0.34)	(-0.12)	(-0.31)	(1.83)	(1.88)	(1.63)	(1.34)
	GARCH				GJR-GARCH				E-GARCH			
θ_0	0.0458***	0.2874***	0.6204***	0.6667***	0.1342***	0.4779***	0.3276***	0.4851***	-0.1081***	-0.1204***	-1.2988***	-1.2856***
	(13.06)	(8.17)	(8.65)	(4.21)	(12.63)	(3.49)	(8.22)	(5.09)	(-9.16)	(-8.35)	(-8.09)	(-7.53)
θ_1/φ_1	0.0747***	0.1510***	0.0242***	0.0650***	0.0258*	0.0061	0.0434***	0.0583***	0.1259***	0.1386***	0.0844***	0.0828**
	(10.58)	(7.50)	(7.34)	(4.13)	(1.93)	(0.38)	(2.86)	(2.93)	(8.53)	(7.96)	(2.72)	(2.52)
γ_1/φ_2					0.2484***	0.0526***	0.1507***	0.1982***	-0.0563***	-0.0743***	-0.1482***	-0.1521***
					(47.68)	(2.58)	(6.15)	(5.76)	(-5.24)	(-6.66)	(-7.46)	(-7.68)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

ω_1/φ_3	0.8545***	0.4602***	0.5692***	0.5604***	0.6622***	0.5468***	0.6955***	0.5675***	0.9860***	0.9839***	0.7953***	0.8010***
	(133.94)	(8.68)	(11.81)	(5.08)	(42.38)	(4.27)	(27.50)	(14.03)	(246.16)	(206.88)	(33.68)	(31.79)
TR_{ETF}	0.3631***	0.3395***	0.4713***	0.3338***	0.3568***	0.2600***	0.4069***	0.3710***	1.1905***	0.9912***	1.2488***	1.0051***
	(30.37)	(14.63)	(38.02)	(3.73)	(36.06)	(2.79)	(36.62)	(19.11)	(22.20)	(14.55)	(25.95)	(14.26)
TR_{mkt}		0.1838***		0.6150***		0.7744***		0.0864**		1.2156***		1.3554***
		(7.97)		(3.06)		(3.74)		(2.51)		(6.27)		(6.97)
VX_{t-1}^2			-0.0529***	-0.0435**			-0.0345***	-0.0497***			0.2004***	0.1980***
			(-5.98)	(-2.38)			(-4.30)	(-2.99)			(7.54)	(7.08)

Note: ( ) represents the Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

GARCH: $\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \omega_1\sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$, GJR-GARCH:$\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \gamma_1\varepsilon_{t-1}^2 DV_{t-1}^- + \omega_1\sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$, E-GARCH: $\ln(\sigma_t^2) = \theta_0 + \varphi_1\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \varphi_2\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \varphi_3\ln(\sigma_{t-1}^2) + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

To further evaluate the response of stock returns to market shocks, the γ_1 coefficient in the GJR-GARCH model and the φ_2 coefficient in the E-GARCH model were analyzed. The γ_1 coefficients are uniformly positive and statistically significant at the 1% level, with values of 0.2484, 0.0526, 0.1507, and 0.1982 in models C5, C6, C7, and C8, respectively. Similarly, the φ_2 coefficients in the E-GARCH models are consistently negative and significant at the 1% level, with values of -0.0563, -0.0743, -0.1482, and -0.1521. These findings corroborate the results from the previous subsection and confirm that the stock return's response to market shocks exhibits negative asymmetry, consistent with prior studies.

In conclusion, the empirical analysis confirms that the growth in ETF trading has a positive impact on stock market volatility, as does increased stock market liquidity. What is more, the stock return's reaction to market shocks exhibits negative asymmetry, meaning that the response to negative shocks is greater than the response to positive ones.

4.2.3 Granger causality test

The empirical evidence indicates that ETFs have a positive influence on stock market volatility. However, Lamoureux and Lastrapes (1990) argue that daily trading volume reflects the confluence of information entering the market, which subsequently impacts market volatility. Additionally, since stock market volatility can also influence trading volume, the causal relationship between these two variables requires further investigation. To explore the potential endogeneity between stock market volatility and trading volume, the Granger causality test is employed, as detailed in the following analysis.

The process of investigating causality involves estimating a Vector Autoregression (VAR) model, identifying the optimal lag structure, and soon after that conducting the Granger causality test. The estimated VAR model is specified as follows:

$$Y_t = \ A_1 Y_{t-1} + \cdots \ldots + A_n Y_{t-n} + \epsilon_t; \qquad Y_t = \begin{bmatrix} \sigma_t \\ TradeVol_{ETF.t} \end{bmatrix} \tag{6}$$

Here, σ_t represents the standard deviation of daily stock returns, calculated using a sample of 30 daily returns. Two measures are used to represent trade volume: the first is the logarithm of dollar volume[6], as suggested by Chordia et al. (2001), and the second is the turnover ratio of ETFs.

[6] Dollar trading volume is a value-based turnover, as defined by Avgouleas and Degiannakis (2009).

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

The optimal lag period is selected based on the Akaike Information Criterion (AIC) or the Schwarz Information Criterion (SIC), and the results of the Granger test are presented in Table 7.

Table 7. Granger test between stock market volatility and ETF turnover

Independent variables	σ_{t-i}, $LnDvol_{t-i}$	σ_{t-i}, $TR_{ETF,t-i}$
Lag period	8	8
AIC	-1.8957#	-1.7443#
SIC	-1.8148	-1.6634
Dep. variable: σ_t		
χ^2	29.7914***	50.9216***
Prob.	0.0002	0.0000
Dep. variable: *TradeVol*		
χ^2	5.9481	18.8628**
Prob.	0.653	0.0156
Lag period	5	3
AIC	-1.8786	-1.7289
SIC	-1.8263#	-1.6956#
Dep. variable: σ_t		
χ^2	19.3589***	42.0296***
Prob.	0.0016	0.0000
Dep. variable: *TradeVol*		
χ^2	8.3976	13.4818***
Prob.	0.1356	0.0037

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

represents the minimum of AIC or SIC.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Table 7 presents the optimal lag lengths determined by the Akaike Information Criterion (AIC) and the Schwarz Information Criterion (SIC). Based on the AIC, the optimal lag length is 8, and the analysis reveals that the dollar trading volume of ETFs Granger-causes stock market volatility, with a χ^2 statistic of 29.7914, which is significant at the 1% level. Conversely, stock market volatility does not Granger-cause the dollar trading volume of ETFs, as indicated by a χ^2 statistic of 5.9481, which fails to achieve statistical significance. In the same manner, under the SIC, where the optimal lag length is identified as 5, the findings are consistent: the dollar trading volume of ETFs Granger-causes stock market volatility, whereas the reverse causality is not supported. Moreover, in examining the Granger causality between stock market volatility and ETF turnover, the results indicate a mutual Granger causality between these two variables, suggesting a bidirectional dynamic in their relationship.

While the turnover of ETFs and stock market volatility exhibit mutual Granger causality, the trading volume of ETFs is specifically identified as a Granger cause of stock market volatility. This observation highlights the potential significance of ETF transactions in shaping the dynamics of stock market volatility, accentuating their influential role within financial markets.

4.2.4 The impact of the creation-redemption mechanism and market turbulence

To confirm the relationship between stock market volatility and ETF turnover, this study applies the methodological framework from Section 3.1 to examine whether the influence of ETFs on stock market volatility differs across market transition period. Using a decade-long sample of 2,444 observations, the analysis divides the period into four distinct sub-periods, facilitating a detailed investigation of temporal variations in ETF impact.

In examining the evolution of ETFs on the Taiwan Stock Exchange, a significant shift occurred on November 2, 2016, when the creation and redemption mechanism transitioned to a dual-track system accommodating both physical and cash creation/redemption processes.[7] As a result, the first sub-period runs from January 1, 2014, to November 1, 2016. The second sub-period is from November 2, 2016 to January 29, 2020. This period includes the Ministry of Health and Welfare in Taiwan issuing an emergency COVID-19 alert on January 23, 2020, extended to January 29, 2020, due to the Chinese New Year holiday (January 22-29). The third sub-period spanning January 30, 2020, to February 23,

[7] Refer to the press release dated December 20, 2016, available on the homepage of Taiwan's Financial Supervisory Commission (FSC). The FSC announced that the cash creation and redemption mechanism was officially permitted to commence on November 2, 2016.
https://www.fsc.gov.tw/ch/home.jsp?id=96&parentpath=0,2&mcustomize=news_view.jsp&dataserno=201612200004&aplistdn=ou=news,ou=multisite,ou=chinese,ou=ap_root,o=fsc,c=tw&dtable=News

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

2022, precedes Russia's invasion of Ukraine on February 24, 2022. The final sub-period, from February 24, 2022, to December 31, 2023, marks the conclusion of the sample period.

To assess whether the impact of ETFs on stock market volatility varies during periods of creation and redemption mechanism transitions and market turbulence, the study re-estimates the GARCH, GJR-GARCH, and E-GARCH models. The empirical results using mean equation (1) are presented in Tables 8 and 9, while those based on mean equation (5) are provided in Table A4 of the Appendix.

Tables 8 and 9 illustrate that the coefficients of ETF turnover remain consistently positive and statistically significant across all models and sub-periods analyzed. Far more significant is the influence of ETFs on stock market volatility intensifies following the implementation of the dual-track creation and redemption mechanism. This likely reflects the increased ease with which investors can exploit deviations between an ETF's net asset value and its market value, thereby amplifying market volatility.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Table 8. The impact of ETFs on stock market volatility across different periods

mean equation: $R_t = \alpha_0 + \alpha_1 R_{t-1} + \varepsilon_t$:							
	θ_0	θ_1/φ_1	γ_1/φ_2	ω_1/φ_3	TR_{ETF}	TR_{mkt}	VX_{t-1}^2
GARCH							
2014.1.1-2016.11.1	0.5510***	0.0021		0.5795***	0.2241*	0.5820*	-0.0361
	(2.60)	(0.29)		(67.67)	(1.74)	(1.69)	(-0.92)
2016.11.2-2020.1.29	-0.0207	0.0460***		0.8075***	0.3278***	0.0240	0.0167
	(-0.41)	(2.59)		(19.98)	(12.89)	(0.47)	(1.55)
2020.1.30-2022.2.23	1.2026***	0.0826***		0.5737***	1.0746**	3.9444***	-0.0434
	(4.36)	(2.93)		(5.62)	(2.48)	(5.83)	(-1.36)
2022.2.24-2023.12.31	0.3226	0.1527***		0.7111***	0.7648***	0.6198***	-0.034
	(1.61)	(3.32)		(12.73)	(9.04)	(3.92)	(-0.95)
2014.1.1-2023.12.31	0.6583***	0.0237***		0.5758***	0.3045***	0.6118***	-0.046**
	(4.17)	(5.02)		(6.38)	(4.40)	(5.23)	(-2.27)
GJR-GARCH							
2014.1.1-2016.11.1	0.5008**	0.0266	-0.0073	0.5763***	0.1496*	0.4465***	-0.0450
	(2.08)	(1.29)	(-0.39)	(3.08)	(1.83)	(2.65)	(-1.63)
2016.11.2-2020.1.29	-0.0043**	-0.0247*	0.1991***	0.7269***	0.3396***	0.0770	0.0182***
	(-2.19)	(-1.83)	(5.46)	(22.01)	(14.18)	(1.58)	(8.23)
2020.1.30-2022.2.23	1.2141***	0.0376	0.0435	0.5751***	1.6411***	2.3860*	-0.0413
	(2.81)	(0.66)	(0.61)	(4.71)	(3.54)	(1.93)	(-1.26)
2022.2.24-2023.12.31	0.5150***	0.0527***	0.1340***	0.7571***	0.7147***	1.0144***	-0.0714**
	(3.03)	(20.12)	(2.97)	(16.50)	(6.88)	(4.56)	(-2.46)
2014.1.1-2023.12.31	0.6446***	0.0150	0.0519***	0.5758***	0.1917***	0.5397***	-0.060***
	(5.55)	(1.45)	(3.89)	(8.97)	(3.64)	(5.15)	(-3.70)
E-GARCH							
2014.1.1-2016.11.1	-1.614***	0.0972	-0.113***	0.7221***	0.9081***	2.3759***	0.2487***
	(-3.86)	(1.34)	(-2.69)	(11.93)	(6.86)	(5.86)	(3.43)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

2016.11.2-2020.1.29	-1.754***	-0.0358	-0.0958**	0.6757***	1.0106***	1.3176***	0.2783***
	(-4.78)	(-0.49)	(-2.33)	(12.67)	(7.79)	(4.25)	(4.55)
2020.1.30-2022.2.23	-1.775***	0.1161	-0.203***	0.7935***	1.3047***	-0.0904	0.2662***
	(-3.44)	(1.16)	(-3.72)	(14.73)	(6.64)	(-0.17)	(3.13)
2022.2.24-2023.12.31	-1.930***	-0.0404	-0.148***	0.7914***	1.2606***	1.2874**	0.3275***
	(-3.44)	(-0.41)	(-2.57)	(15.24)	(5.33)	(2.20)	(3.39)
2014.1.1-2023.12.31	-1.296***	0.0817**	-0.151***	0.7994***	1.0047***	1.3696***	0.2000***
	(-7.51)	(2.52)	(-7.63)	(31.47)	(14.16)	(6.98)	(7.07)

Note: ( ) : Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

GARCH: $\sigma_t^2 = \theta_0 + \theta_1 \varepsilon_{t-1}^2 + \omega_1 \sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$, GJR-GARCH: $\sigma_t^2 = \theta_0 + \theta_1 \varepsilon_{t-1}^2 + \gamma_1 \varepsilon_{t-1}^2 DV_{t-1}^- + \omega_1 \sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$, E-GARCH: $\ln(\sigma_t^2) = \theta_0 + \varphi_1 \left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \varphi_2 \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \varphi_3 \ln(\sigma_{t-1}^2) + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2$.

Table 9. Trends in the coefficients of variance equations during transition periods

	GARCH			GJR-GARCH			E-GARCH		
Period	Cash C. & R.	COVID-19	Russia-Ukraine War	Cash C. & R.	COVID-19	Russia-Ukraine War	Cash C. & R.	COVID-19	Russia-Ukraine War
θ_1/φ_1	+	+	+	−	+	+	−	+	−
γ_1/φ_2				+	−	+	−	+	−
ω_1/φ_3	+	−	+	+	−	+	−	+	−
TR_{ETF}	+	+	−	+	+	−	+	+	−
TR_{mkt}	−	+	−	−	+	−	−	−	+
VX_{t-1}^2	+	−	+	+	−	−	+	−	+

Note: 1. Cash C.&R. represents cash creation and redemption, and it was introduced on November 1, 2016.

2. +: The trend of the coefficients is increasing. −: The trend of the coefficients is decreasing.

During the COVID-19 period, the coefficients of ETF turnover are significantly higher than in earlier periods, manifesting themselves with the pandemic's pronounced impact on

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

market turbulence. This can be attributed to Authorized Participants (APs) and arbitrageurs ceasing trading during periods of extreme volatility, which reduced market liquidity and further exacerbated stock market volatility. These findings suggest that even at similar levels of ETF activity, the associated market volatility was substantially elevated, underscoring the heightened fragility of the stock market during this unprecedented crisis.

Although the Russia-Ukraine War also disrupted the stock market, as shown by the elevated coefficients of ETF turnover (0.7648, 0.7147, 1.2606) compared to those for the full sample period (0.3045, 0.1917, 1.0047), its impact was less severe than that of COVID-19. This suggests that while the war generated market volatility, it did not reach the same magnitude of disruption as the pandemic.

Analyzing the evolution of coefficients for stock market turnover reveals a consistently positive impact on volatility, except during the COVID-19 period in the E-GARCH model, where the relationship is statistically insignificant. Rather noticeably, a divergence per se emerges between the effects of stock market turnover and ETF turnover. As the ETF cash creation and redemption mechanism gains prominence, the influence of stock market turnover on volatility diminishes, while ETFs' impact intensifies significantly. This shift is evident in the GARCH model, where ETF coefficients range from 0.2241 to 0.3278, while stock market turnover coefficients decline from 0.582 to 0.024. This mechanism attracts investors toward arbitrage activities, amplifying stock market volatility.

During the pandemic crisis, stock market turnover had an increasing impact on volatility across all models except E-GARCH, where its effect was insignificant. This growth, more significant than that of ETFs (with a greater coefficient for market turnover), indicates that COVID-19 affected the broader stock market more than ETFs. Conversely, during the Russia-Ukraine war, the influence of stock market turnover declined in most models, following the trend in ETFs, except in the E-GARCH model. These patterns put emphasis on varying market responses to crises under different conditions.

Furthermore, Stock returns are more sensitive to negative shocks, as shown by the consistently negative and significant φ_2 coefficients in the E-GARCH model. However, the degree of this asymmetry varies markedly during transitional periods.

During the phase when the creation and redemption mechanisms for ETFs operated under a dual-track system, the asymmetry of negative responses diminished, with coefficients varying from -0.113 to -0.0958. This reduction can be attributed to the dual-track mechanism providing a more accessible avenue for trading ETFs in the primary market, which mitigates the downward pressure on ETF prices. Conversely, during the COVID-19 pandemic, the asymmetry of responses intensified. The unprecedented and

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

severe negative shocks associated with the pandemic amplified the market's adverse reactions.

Post-pandemic, the Russia-Ukraine war reduced market response asymmetry, indicating less severe shocks compared to COVID-19. ETFs maintained a consistent positive relationship with stock market volatility, playing a prominent part during the economic and geopolitical turbulence.

In conclusion, ETFs and stock market turnover consistently drive stock market volatility during transitional periods. The influence of ETF turnover has intensified with the adoption of cash-based creation and redemption mechanisms and during the COVID-19 pandemic but weakened during the Russia-Ukraine war. This heightened impact, particularly during dual mechanisms and market turbulence, corresponds to previous findings linking it to the behaviors of arbitrageurs and APs. In contrast, the effect of stock market turnover diminishes under cash-based mechanisms and during the war but surges drastically during the pandemic. Moreover, the asymmetric response of volatility to negative market shocks remains robust across all periods, putting emphasis on its enduring relevance in diverse market conditions.

4.2.5 Influence of the number of listed ETFs on stock market volatility

Since the introduction of ETFs to the Taiwan Stock Exchange in 2003, their growth remained modest until 2017 but surged notably between 2021 and 2023. This raises the question of whether the rising number of ETFs influences stock market volatility. To explore this, the study extends the empirical model by incorporating the number of ETFs as a regressor, with the corresponding variance equations presented.

GARCH: $$\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \omega_1\sigma_{t-1}^2 + \beta_1 TradeVol_{ETF.t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 + \beta_4 NO_{ETF,t} \quad (7)$$

GJR-GARCH: $$\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \gamma_1\varepsilon_{t-1}^2 DV_{t-1}^- + \omega_1\sigma_{t-1}^2 + \beta_1 TradeVol_{ETF.t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 + \beta_4 NO_{ETF,t} \quad (8)$$

E-GARCH: $$\ln(\sigma_t^2) = \theta_0 + \varphi_1\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \varphi_2\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \varphi_3\ln(\sigma_{t-1}^2) + \beta_1 TradeVol_{ETF.t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 + \beta_4 NO_{ETF,t} \quad (9)$$

ETF trading volume ($TradeVol_{ETF.t}$) is measured using two approaches: $TR_{ETF,t}$ and $DVOL_{ETF,t}$, where $DVOL_{ETF,t}$ represents the logarithmic difference of dollar trading volume. Additionally, $NO_{ETF,t}$ denotes the number of listed ETFs. The empirical results

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

are shown in Table 10.

Table 10 shows that the effects of ETFs and stock market turnover on stock volatility remain positive and statistically significant across various variance models, consistently at the 1% level. The negative asymmetry in response to market shocks is confirmed in both the GJR-GARCH and E-GARCH models, as indicated by the coefficients $\gamma_1 > 0$ and $\varphi_2 < 0$. This result holds even when dollar trading volume is used as a substitute for turnover.

The influence of the number of ETFs on stock market volatility is consistently negative across all models, though some coefficients lack statistical significance. This suggests that a growing number of ETFs tends to reduce market volatility. The underlying mechanism stems from the diversification of ETFs introduced in Taiwan, particularly those tied to domestic equities. The broader distribution of newly launched ETFs across various categories mitigates market risk, as the systematic risk of these new ETFs is lower compared to the earlier ones, thereby reducing the overall systematic risk of the ETF portfolio.

Table 10. The impact of the number of listed ETFs on stock market volatility, regressed against ETF turnover and dollar volume, respectively.

	TR_{ETF}	$DVOL_{ETF}$	TR_{ETF}	$DVOL_{ETF}$	TR_{ETF}	$DVOL_{ETF}$
	GARCH		GJR-GARCH		E-GARCH	
Incorporated with mean equation (1)						
θ_0	0.7268***	0.7661***	0.7134***	0.7008***	-1.2969***	-0.8218***
	(4.94)	(5.02)	(6.49)	(12.21)	(-7.44)	(-6.71)
θ_1/φ_1	0.0706***	0.0356***	0.0169	0.0397**	0.0816**	0.0648**
	(5.80)	(4.55)	(1.50)	(2.15)	(2.50)	(2.40)
γ_1/φ_2			0.0524***	0.0916***	-0.1506***	-0.1555***
			(3.67)	(3.56)	(-7.61)	(-8.98)
ω_1/φ_3	0.5765***	0.5828***	0.5809***	0.5752***	0.7992***	0.8661***
	(9.02)	(5.29)	(8.21)	(9.50)	(31.41)	(45.00)
TR_{ETF}	0.2989***	0.4002***	0.2182***	0.2210***	1.0049***	1.0138***
	(6.15)	(4.92)	(4.62)	(4.72)	(14.11)	(13.36)
TR_{mkt}	0.4833***	0.7097***	0.5841***	0.5423***	1.3693***	1.6452***

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

	(6.90)	(5.75)	(8.25)	(9.06)	(6.96)	(7.73)
VX_{t-1}^2	-0.0418**	-0.0282	-0.0389***	-0.0392***	0.2007***	0.1243***
	(-2.03)	(-1.41)	(-7.37)	(-2.88)	(6.99)	(5.99)
NO_{ETF}	-0.0687***	-0.0519	-0.0687***	-0.0709***	-0.0010	-0.0013
	(-21.84)	(-1.14)	(-2.67)	(-2.77)	(-0.06)	(-0.11)
Incorporated with mean equation (5)						
θ_0	0.6591***	0.7624***	0.6954***	0.7561***	-1.2888***	-0.8167***
	(8.18)	(5.97)	(9.76)	(5.64)	(-7.53)	(-6.78)
θ_1/φ_1	0.0884***	0.0457***	0.0560***	-0.0245***	0.0826**	0.0649**
	(5.73)	(4.79)	(4.12)	(-4.31)	(2.49)	(2.34)
γ_1/φ_2			0.0670***	0.0789***	-0.1520***	-0.1557***
			(2.89)	(5.79)	(-7.66)	(-8.97)
ω_1/φ_3	0.5674***	0.5810***	0.5800***	0.5824***	0.8007***	0.8669***
	(8.09)	(5.48)	(10.34)	(7.67)	(31.72)	(45.63)
TR_{ETF}	0.3288***	0.4242***	0.2619***	0.5360***	1.0055***	1.0164***
	(6.09)	(5.05)	(5.44)	(6.95)	(14.20)	(13.47)
TR_{mkt}	0.5767***	0.6464***	0.6335***	0.4410**	1.3551***	1.6041***
	(6.51)	(3.83)	(8.23)	(2.06)	(6.96)	(7.61)
VX_{t-1}^2	-0.0349***	-0.0287	-0.0446***	-0.0274	0.1989***	0.1236***
	(-3.47)	(-1.24)	(-3.99)	(-1.26)	(7.02)	(6.04)
NO_{ETF}	-0.0071***	-0.0534	-0.0076***	-0.0544	-0.0001	-0.0019
	(-5.40)	(-1.19)	(-6.96)	(-1.54)	(-0.11)	(-0.17)

Note: ( ) represents the Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

GARCH: $\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \omega_1\sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 + \beta_4 NO_{ETF,t}$, GJR-GARCH:$\sigma_t^2 = \theta_0 + \theta_1\varepsilon_{t-1}^2 + \gamma_1\varepsilon_{t-1}^2 DV_{t-1}^- + \omega_1\sigma_{t-1}^2 + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 + \beta_4 NO_{ETF,t}$,

E-GARCH: $\ln(\sigma_t^2) = \theta_0 + \varphi_1\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \varphi_2\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \varphi_3\ln(\sigma_{t-1}^2) + \beta_1 TR_{ETF,t} + \beta_2 TR_{mkt,t} + \beta_3 VX_{t-1}^2 + \beta_4 NO_{ETF,t}$.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

5. CONCLUSION

The rapid growth of exchange-traded fund (ETF) trading reflects smaller investors' adoption of diversified strategies aligned with the efficient market hypothesis (EMH) and the appeal of ETFs' accessibility, low costs, and tax advantages. While high-frequency traders, including noisy traders, may disrupt price discovery, authorized participants (APs) and arbitrageurs enhance it by aligning ETF prices with their underlying assets through creation, redemption, and arbitrage. Consequently, ETF trading may increase stock return volatility, though the liquidity they provide helps stabilize abrupt price movements, underscoring the intricate interplays among liquidity, price discovery, and volatility in financial markets.

Thus, this paper employs three GARCH models to investigate the effect of ETF trading upon stock market volatility by using data from the Taiwan Stock Exchange (TWSE). The empirical results demonstrate that stock return volatility is influenced by ETF trading activity, represented by the turnover ratio, as well as by stock market liquidity, designated by the stock index turnover ratio. Both factors positively affect stock return volatility. This coincides with findings by Ben-David et al. (2018), suggesting that higher ETF turnover, often driven by high-frequency traders, leads to increased volatility. Increased stock market liquidity in the meantime raises stock volatility.

Furthermore, the response of the stock market to external shocks is analyzed through the coefficients of the dummy variable in the GJR-GARCH model (γ_1) and the lagged error term in the E-GARCH model (φ_2). The results reveal that the stock market in Taiwan exhibits a negatively asymmetric reaction, responding more strongly to "bad" news than to "good" news.

This paper truly examines the effect of ETFs on stock market volatility, focusing on the transition of the creation and redemption mechanism and the influence of market turbulence. The analysis reveals that the shift to a cash-based creation and redemption mechanism amplifies the impact of ETFs on market volatility. This finding supports previous research, which suggests that cash-based mechanisms facilitate speculative trading, thereby increasing market fluctuations.

In the context of market turbulence, the results demonstrate that the impact of ETFs on stock market volatility intensifies. This is consistent with the findings of Ben-David et al. (2017), who argue that diminished information reliability during turbulent periods leads to a withdrawal of trading activity by authorized participants (APs) and arbitrageurs, further exacerbating volatility. Moreover, the study confirms the presence of a negative asymmetric

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

response in stock return volatility to market shocks. These results provide robust validation of the study's key conclusions.

Besides, this study reveals a negative relationship between the number of ETFs and stock market volatility, suggesting that an increase in ETFs contributes to market stability. This stabilizing effect can be attributed to the diversification introduced by ETFs in Taiwan, particularly those linked to domestic equities. The broader distribution of newly launched ETFs across diverse categories helps mitigate market risk, as these newer ETFs exhibit lower systematic risk compared to earlier ones. As a result, the overall systematic risk of the ETF portfolio declines, further enhancing market stability.

These findings suggest that the expansion of ETF trading activity is associated with a corresponding increase in stock return volatility, carrying significant implications for investment strategies. From a perspective of investment management, investors should closely monitor ETF trading activity, as it influences stock volatility and, consequently, the required rate of return. Adopting appropriate investment strategies will be essential to adapt to the evolving dynamics of ETF trading.

In a nutshell, the speed at which discrepancies between ETF prices and the value of their underlying assets are resolved depends on the actions of key market participants, such as authorized participants (APs), high-frequency arbitrageurs, and speculators. Regulators and investors must carefully consider these participants' responses to new ETF trading mechanisms or market turbulence, as their behavior can influence price discovery efficiency and amplify stock market volatility. This reflects the U.S. Securities and Exchange Commission's (SEC) regulations, which permit trading halts for individual securities and ETFs during periods of extreme volatility to maintain market stability (see Ben-David et al., 2017).

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

REFERENCES

Alberg, D., Shalit, H., & Yosef, R. (2008). Estimating Stock Market Volatility Using Asymmetric GARCH Models. *Applied Financial Economics*, 18, 1201-1208.

https://doi.org/10.1080/09603100701604225

Anadu, K., Kruttli, M., McCabe, P., & Osambela, E. (2020). The Shift from Active to Passive Investing: Risks to Financial Stability? *Financial Analysts Journal*, 76(4), 23–39. https://doi.org/10.1080/0015198X.2020.1779498

Aragon G. O., & Strahan P. E. (2012). Hedge Funds as Liquidity Providers: Evidence from the Lehman Bankruptcy. *Journal of Financial Economics*, 103(3), 570-587.

https://doi.org/10.1016/j.jfineco.2011.10.004

Avgouleas, E., & Degiannakis, S. (2009). Trade Transparency and Trading Volume: The Impact of the Financial Instruments Markets Directive on the Trading Volume of EU Equity Markets. *International Journal of Financial Markets and Derivatives*, l, 96-123. https://doi.org/10.1504/IJFMD.2009.028947

Ben-David, I., Franzoni, F., & Moussawi, R. (2012). Hedge Funds Stock Trading in the Financial Crisis of 2007-2009. *Review of Financial Studies*, 25(1), 1-54.

https://doi.org/10.1093/rfs/hhr114

Ben-David, I., Franzoni, F., & Moussawi, R. (2017). Exchange-Traded Funds. *Annual Review of Financial Economics*, 9, 169–189. https://www.jstor.org/stable/26774086

Ben-David, I., Franzoni, F., & Moussawi, R. (2018). Do ETFs Increase Volatility? *The Journal of Finance*, 73, 2471-2535. https://doi.org/10.1111/jofi.12727

Bollerslev, T. (1986). Generalized Autoregressive Conditional Heteroscedasticity. *Journal of Econometrics*, 31, 307-327. https://doi.org/10.1016/0304-4076(86)90063-1

Bollerslev, T. (1987). A Conditional Heteroskedastic Time Series Model for Speculative Prices and Rates of Return. *Review of Economics and Statistics*, 69, 542-547.

https://doi.org/10.2307/1925546

Broman M. S. (2016). Liquidity, Style Investing and Excess Comovement of Exchange-Traded Fund Returns. *Journal of Financial Markets*, 30, 27-53 .

https://doi.org/10.1016/j.finmar.2016.05.002

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Broman, M. S. (2022). Naïve Style-Level Feedback Trading in Passive Funds. *Journal of Financial and Quantitative Analysis*, 57(3), 1083–1114.

https://doi.org/10.1017/S0022109021000247

Brown, S. J. (2020). The Efficient Market Hypothesis, the Financial Analysts Journal, and the Professional Status of Investment Management. *Financial Analysts Journal*, 76(2), 5-14. https://doi.org/10.1080/0015198X.2020.1734375

Brown, D. C., Davies S. W., & Ringgenberg M. C. (2021). ETF Arbitrage, Non-Fundamental Demand, and Return Predictability. *Review of Finance*, 937–972.

https://doi.org/10.1093/rof/rfaa027

Chabakauri, G., & Rytchkov, O. (2021). Asset pricing with index investing, *Journal of Financial Economics*, 141(1), 195-216.https://doi.org/10.1016/j.jfineco.2020.06.023.

Chordia, T., Subrahmanyam, A., & Anshuman, V. R. (2001). Trading Activity and Expected Stock Returns. *Journal of Financial Economics*, 59, 3-32.

https://doi.org/10.1016/S0304-405X(00)00080-5

Datar, V. T., Naik, N. Y. & Radcliffe, R. (1998). Liquidity and Stock Returns: an Alternative Test. *Journal of Financial Markets*, 1, 203-219.

https://doi.org/10.1016/S1386-4181(97)00004-9

Dedi, L., Yavas, B. F., & McMillan, D. (2016). Return and Volatility Spillovers in Equity Markets: An Investigation Using Various GARCH Methodologies. *Cogent Economics & Finance*, 4(1), 1-18. https://doi.org/10.1080/23322039.2016.1266788

Duan, C.-W., & Lin, J.-C. (2014). The Predictive Power of Volatility Models: Evidence from the ETF Market. *Investment Management and Financial Innovations*, 11(2), 100-110.

Elton, E. J., Gruber, M. J. & Souza, A. (2019). Passive Mutual Funds and ETFs: Performance and Comparison. *Journal of Banking and Finance*, 106, 265-275.

https://doi.org/10.1016/j.jbankfin.2019.07.004

Engle, R. F. (1982). Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of U.K. Inflation. *Econometrica*, 50, 987-1008.

https://doi.org/10.2307/1912773

Engle, R. F., & Ng, V. K. (1993). Measuring and Testing the Impact of News on Volatility.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

The Journal of Finance, 48(5), 1749–1778. https://doi.org/10.2307/2329066

Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, 25(2), 383–417. https://doi.org/10.2307/2325486

Financial Supervisory Commission. (2016). *Continuously Promote the Development of Diversified ETF Products* (press release dated December 20, 2016), Financial Supervisory Commission, R.O.C.

https://www.fsc.gov.tw/ch/home.jsp?id=96&parentpath=0,2&mcustomize=news_view.jsp&dataserno=201612200004&aplistdn=ou=news,ou=multisite,ou=chinese,ou=ap_root,o=fsc,c=tw&dtable=News

Glosten, L., Nallareddy S., & Zou Y. (2021). ETF Activity and Informational Efficiency of Underlying Securities. *Management Science*, 67(1), 22–47.

https://doi.org/10.1287/mnsc.2019.3427

Guzman, S., Peteul, J., & Rezaee, A. (2022). How Have ETFs Changed Market Macro Efficiency and Risk Structure? *Journal of Portfolio Management*, 48(7), 156–174. https://doi.org/10.3905/jpm.2022.1.361

Ibbotson, R. G., Chen, Z., Kim, D. Y. J., & Hu, W. Y. (2013). Liquidity as an Investment Style. *Financial Analysts Journa*l, 69(3), 30–44.

http://dx.doi.org/10.2139/ssrn.1817889

Kapusuzoglu, A., & Ceylan, N. B. (2018). Trading Volume, Volatility and GARCH Effects in Borsa Istanbul. In *Strategic Design and Innovative Thinking in Business Operations*, Springer: Cham, Switzerland, 333–347. http://dx.doi.org/10.1007/978-3-319-77622-4_17

Karmaziene, E., & Sokolovski, V. (2022). Short-Selling Equity Exchange Traded Funds and Its Effect on Stock Market Liquidity. *Journal of Financial and Quantitative Analysis*, 57(3), 923–956. https://doi.org/10.1017/S0022109021000181

Kim, M., & Kim, J.-S. (2023). The Effect of ETFs on Underlying Stock Markets: Evidence from Korea. Available at SSRN: https://ssrn.com/abstract=4343661 or

http://dx.doi.org/10.2139/ssrn.4343661

Lamoureux, C. G. & Lastrapes, W. D. (1990). Heteroskedasticity in Stock Return Data: Volume versus GARCH Effects. *Journal of Finance*, 45(1), 221-229.

https://doi.org/10.1111/j.1540-6261.1990.tb05088.x

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Lettau, M., & Madhavan, A. (2018). Exchange-Traded Funds 101 for Economists. *The Journal of Economic Perspectives*, 32(1), 135–154.

https://doi.org/10.1257/jep.32.1.135

Li, M. & Zhao, X. (2014). Impact of Leveraged ETF Trading on the Market Quality of Component Stocks. *The North American Journal of Economics and Finance*, 28, 90-108. https://doi.org/10.1016/j.najef.2014.02.001

Madhavan A. N. (2016). *Exchange-Traded Funds and the New Dynamics of Investing*. Oxford, UK: Oxford Univ. Press.

https://doi.org/10.1093/acprof:oso/9780190279394.001.0001

Marta, T. & Riva, F. (2025) Do ETFs Increase the Comovements of Their Underlying Assets? Evidence from a Switch in ETF Replication Technique. *Journal of Banking & Finance*, 170, 107333. https://doi.org/10.1016/j.jbankfin.2024.107333

Nelson, D. B. (1991). Conditional Heteroskedasticity in Asset Returns: A New Approach. *Econometrica*, 59(2), 347–370. https://doi.org/10.2307/2938260

Qin, N., & Singal, V. (2015). Indexing and Stock Price Efficiency. *Financial Management*, 44(4), 875–904. http://www.jstor.org/stable/24736544

Rossi, G. D. & Steliaros, M. (2022). The Shift from Active to Passive and its Effect on Intraday Stock Dynamics. *Journal of Banking and Finance*, (143), 106595.

https://doi.org/10.1016/j.jbankfin.2022.106595

Tkac, P. A. (1999). A Trading Volume Benchmark: Theory and Evidence. *The Journal of Financial and Quantitative Analysis*, 34(1), 89-114. https://doi.org/10.2307/2676247

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

APPENDIX

Table A1. Market value and trading value ratios of Taiwan's domestic equity ETFs from 2014 to 2023. Unit: %

ETF / Year	0050	0051	0052	0053	0055	0056	0057	6203	6204	6208
Market value ratio										
2014	0.2929	0.0022	0.0013	0.0026	0.0077	0.0166	0.0095	0.0629	0.0019	0.0005
2015	0.3139	0.0019	0.0013	0.0019	0.0056	0.0225	0.0018	0.0321	0.0011	0.0005
2016	0.2536	0.0018	0.0013	0.0015	0.0038	0.0317	0.0013	0.0011	0.0009	0.0006
2017	0.1486	0.0013	0.0011	0.0010	0.0025	0.0292	0.0008	0.0009	0.0005	0.0030
2018	0.1811	0.0009	0.0011	0.0007	0.0020	0.0370	0.0008	0.0009	0.0005	0.0036
2019	0.2621	0.0009	0.0026	0.0007	0.0022	0.0688	0.0008	0.0011	0.0005	0.0050
2020	0.2631	0.0010	0.0059	0.0007	0.0018	0.1551	0.0009	0.0014	0.0003	0.0111
2021	0.3123	0.0011	0.0131	0.0007	0.0016	0.1624	0.0009	0.0014	0.0003	0.0264
2022	0.4745	0.0016	0.0124	0.0006	0.0037	0.2883	0.0005	0.0014	0.0003	0.0665
2023	0.5769	0.0021	0.0110	0.0006	0.0034	0.4201	0.0003	0.0014	0.0003	0.1072
Average	0.3079	0.0015	0.0051	0.0011	0.0034	0.1232	0.0018	0.0105	0.0007	0.0224
Trading value ratio										
2014	0.7433	0.0015	0.0007	0.0021	0.0141	0.0326	0.0048	0.0463	0.0125	0.0007
2015	0.7616	0.0012	0.0008	0.0013	0.0141	0.0444	0.0032	0.0007	0.0009	0.0006
2016	0.8476	0.0010	0.0005	0.0010	0.0058	0.0884	0.0011	0.0004	0.0011	0.0002
2017	0.2792	0.0014	0.0005	0.0007	0.0019	0.0474	0.0006	0.0002	0.0005	0.0005
2018	0.4960	0.0006	0.0028	0.0004	0.0022	0.0855	0.0008	0.0003	0.0005	0.0016
2019	0.5457	0.0007	0.0054	0.0004	0.0019	0.2094	0.0013	0.0003	0.0009	0.0149
2020	0.4827	0.0009	0.0217	0.0006	0.0006	0.3099	0.0019	0.0003	0.0012	0.0357

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

2021	0.3632	0.0010	0.0326	0.0004	0.0020	0.1764	0.0008	0.0001	0.0007	0.0456
2022	0.6168	0.0010	0.0214	0.0002	0.0058	0.3275	0.0020	0.0002	0.0010	0.1061
2023	0.4665	0.0015	0.0132	0.0002	0.0018	0.3524	0.0003	0.0001	0.0009	0.1021
Average	0.5603	0.0011	0.0100	0.0007	0.0050	0.1674	0.0017	0.0049	0.0020	0.0308

Note: The market value ratio is the market value of the fund relative to the stock market value, while the trading value ratio is the trading value of the fund relative to the stock market trading value.

The following Taiwan domestic equity ETFs were available during the period from June 30, 2003, to July 17, 2012: Yuanta/P-shares Taiwan Top 50 ETF (0050), Yuanta/P-shares Taiwan Mid-Cap 100 ETF (0051), Fubon Taiwan Technology Tracker Fund (0052), Yuanta/P-shares Taiwan Electronics Tech ETF (0053), Yuanta/P-shares MSCI Taiwan Financials ETF (0055), Yuanta/P-shares Taiwan Dividend Plus ETF (0056), Fubon MSCI® Taiwan ETF (0057), Yuanta/P-shares MSCI Taiwan ETF (006203), Sinopac TAIEX ETF (006204), and Fubon FTSE TWSE Taiwan 50 ETF (006208).

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Table A2. The comparison of coefficients in the variance equation between the ETFs 0050 and 0056 incorporated with mean equation (1)

The coefficients of variance equation in GARCH model								
Condition	TR_{ETF} (C1)		TR_{ETF}, TR_{mkt} (C2)		TR_{ETF}, VX (C3)		TR_{ETF}, TR_{mkt}, VX, (C4)	
ETF	**0050**	**0056**	**0050**	**0056**	**0050**	**0056**	**0050**	**0056**
θ_0	0.074^{***}	0.049^{***}	0.696^{***}	0.070^{***}	0.289^{***}	0.022	0.730^{***}	0.521^{***}
	(14.31)	(6.01)	(4.34)	(6.30)	(6.70)	(1.24)	(4.24)	(6.45)
θ_1	0.078^{***}	0.094^{***}	0.003	0.092^{***}	0.101^{***}	0.082^{***}	0.020^{***}	0.189^{***}
	(8.51)	(8.10)	(1.17)	(7.28)	(9.09)	(7.40)	(4.52)	(3588.9)
ω_1	0.809^{***}	0.837^{***}	0.572^{***}	0.805^{***}	0.781^{***}	0.821^{***}	0.564^{***}	0.555^{***}
	(64.29)	(43.38)	(6.12)	(32.83)	(38.81)	(36.56)	(5.19)	(19.06)
TR_{ETF}	0.332^{***}	0.257^{***}	0.729^{***}	0.231^{***}	0.370^{***}	0.262^{***}	0.440^{***}	0.278^{***}
	(29.14)	(34.03)	(6.82)	(15.44)	(35.8)	(26.05)	(5.35)	(22.88)
TR_{mkt}			0.552^{***}	0.404^{***}			0.550^{***}	0.428^{***}
			(4.78)	(16.65)			(3.15)	(8.78)
VX_{t-1}^2					-0.037^{***}	0.008^{**}	-0.040^{**}	-0.056^{***}
					(-5.75)	(2.01)	(-2.06)	(-3.94)
The coefficients of variance equation in GJR-GARCH model								
θ_0	0.148^{***}	0.067^{***}	0.481^{***}	0.464^{***} (4.78)	0.330^{***}	-0.059^{*}	0.515^{***}	0.638^{***}
	(645.32)	(7.56)	(3.45)		(9.57)	(-1.90)	(5.81)	(17.14)
θ_1	-0.001	0.038^{***}	-0.003	-0.001	0.012	-0.0005	0.030^{*}	0.018^{*}
	(-0.13)	(2.89)	(-0.18)	(-0.05)	(1.16)	(-0.05)	(1.82)	(1.82)
γ_1	0.179^{***}	0.130^{***}	0.063^{***}	0.064^{***}	0.191^{***}	0.162^{***}	0.212^{***}	0.051^{***}
	(10.18)	(6.73)	(3.11)	(3.27)	(7.66)	(8.41)	(7.17)	(4.31)
ω_1	0.691^{***}	0.799^{***}	0.557^{***}	0.556^{***}	0.645^{***}	0.762^{***}	0.579^{***}	0.576^{***}
	(67.64)	(37.35)	(4.36)	(6.12)	(20.79)	(36.05)	(19.13)	(12.10)
TR_{ETF}	0.329^{***}	0.247^{***}	0.241^{***}	0.326^{***}	0.355^{***}	0.244^{***}	0.363^{***}	0.226^{***}
	(31.54)	(31.30)	(2.78)	(4.67)	(29.69)	(20.88)	(26.93)	(6.90)
TR_{mkt}			0.768^{***}	0.899^{***}			0.027^{***}	0.633^{***}
			(4.35)	(5.87)			(18.39)	(16.81)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

VX_{t-1}^2					-0.028***	0.030***	-0.054***	-0.061***
					(-8.71)	(4.17)	(-3.53)	(-7.33)
The coefficients of variance equation in E-GARCH model								
θ_0	-0.108***	-0.116***	-0.119***	-0.146***	-0.823***	-1.482***	-0.966***	-1.384***
	(-9.24)	(-9.34)	(-8.08)	(-8.91)	(-7.76)	(-8.68)	(-6.99)	(-7.67)
φ_1	0.122***	0.136***	0.134***	0.166***	0.067***	0.096***	0.073**	0.086***
	(8.20)	(8.98)	(7.49)	(8.63)	(2.66)	(3.20)	(2.43)	(2.61)
φ_2	-0.063***	-0.069***	-0.082***	-0.098***	-0.130***	-0.158***	-0.144***	-0.162***
	(-5.54)	(-6.82)	(-6.85)	(-8.30)	(-7.79)	(-7.91)	(-8.08)	(-8.36)
φ_3	0.981***	0.982***	0.980***	0.973***	0.862***	0.753***	0.844***	0.777***
	(233.51)	(206.95)	(201.63)	(148.82)	(51.41)	(28.98)	(40.59)	(28.14)
TR_{ETF}	1.201***	0.730***	1.020***	0.477***	1.249***	0.772***	1.017***	0.471***
	(23.34)	(22.42)	(15.70)	(9.55)	(27.02)	(24.55)	(15.40)	(8.96)
TR_{mkt}			1.321***	1.941***			1.500***	2.154***
			(6.77)	(11.00)			(8.12)	(11.74)
VX_{t-1}^2					0.124***	0.232***	0.146***	0.215***
					(6.85)	(7.99)	(6.49)	(7.13)

Note: ( ) represents the Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

The GARCH, GJR-GARCH, and E-GARCH models correspond to equations (2), (3), and (4), respectively.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Table A3. The comparison of coefficients in the variance equation between the ETFs 0050 and 0056 incorporated with mean equation (5)

The coefficients of variance equation in GARCH model								
Condition	TR_{ETF} (C1)		TR_{ETF}, TR_{mkt} (C2)		TR_{ETF}, VX (C3)		TR_{ETF}, TR_{mkt}, VX, (C4)	
ETF	**0050**	**0056**	**0050**	**0056**	**0050**	**0056**	**0050**	**0056**
θ_0	0.074***	0.050***	0.267***	0.295***	0.675***	0.022	0.742***	0.656***
	(14.19)	(5.95)	(8.70)	(7.47)	(11.89)	(1.25)	(3.38)	(11.87)
θ_1	0.078***	0.094***	0.151***	0.120***	0.034***	0.082***	0.040***	0.077***
	(8.43)	(8.04)	(7.73)	(8.43)	(6.75)	(7.36)	(3.45)	(6.23)
ω_1	0.809***	0.837***	0.477***	0.508***	0.562***	0.821***	0.567***	0.566***
	(62.41)	(43.05)	(9.78)	(9.64)	(8.99)	(36.31)	(4.67)	(8.95)
TR_{ETF}	0.332***	0.257***	0.321***	0.266***	0.561***	0.262***	0.526***	0.433***
	(26.90)	(32.91)	(26.69)	(15.64)	(36.48)	(24.77)	(5.17)	(28.32)
TR_{mkt}			0.104***	0.518***			0.538**	0.736***
			(12.20)	(14.13)			(2.39)	(19.37)
VX_{t-1}^2					-0.050***	0.008**	-0.034	-0.046***
					(-32.14)	(2.00)	(-1.30)	(-7.54)
The coefficients of variance equation in GJR-GARCH model								
θ_0	0.144***	0.068***	0.481***	0.459*** (5.23)	0.387***	-0.058*	0.623***	0.612***
	(18.86)	(7.41)	(3.29)		(5.86)	(-1.84)	(6.30)	(7.35)
θ_1	-0.003	0.038***	0.006	-0.023***	0.012	-0.0003	0.061***	0.035**
	(-0.44)	(2.90)	(0.43)	(-3.77)	(1.23)	(-0.03)	(4.91)	(2.25)
γ_1	0.204***	0.131***	0.046**	0.139***	0.203***	0.162***	0.077***	0.100***
	(13.81)	(6.53)	(2.44)	(5.45)	(12.17)	(8.28)	(3.31)	(4.59)
ω_1	0.690***	0.798***	0.550***	0.544***	0.633***	0.762***	0.580***	0.561***
	(54.69)	(36.46)	(4.07)	(6.40)	(21.17)	(35.70)	(10.62)	(11.35)
TR_{ETF}	0.320***	0.249***	0.218***	0.333***	0.368***	0.244***	0.155***	0.340***
	(28.50)	(30.98)	(2.65)	(5.06)	(27.28)	(19.95)	(3.56)	(30.63)
TR_{mkt}			0.792***	0.850***			0.434***	0.547***
			(4.29)	(4.65)			(4.58)	(12.59)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

VX_{t-1}^2					-0.036***	0.030***	-0.068***	-0.056***
					(-3.59)	(4.10)	(-4.31)	(-4.79)
The coefficients of variance equation in E-GARCH model								
θ_0	-0.109***	-0.118***	-0.119***	-0.147***	-0.827***	-1.439***	-0.967***	-1.377***
	(-9.03)	(-9.28)	(-7.92)	(-8.97)	(-7.80)	(-8.80)	(-7.01)	(-7.72)
φ_1	0.123***	0.139***	0.135***	0.167***	0.068***	0.097***	0.073**	0.088***
	(8.07)	(8.92)	(7.39)	(8.66)	(2.61)	(3.22)	(2.41)	(2.65)
φ_2	-0.064***	-0.069***	-0.082***	-0.097***	-0.132***	-0.164***	-0.144***	-0.164***
	(-5.56)	(-6.78)	(-6.84)	(-8.19)	(-7.83)	(-8.26)	(-8.09)	(-8.42)
φ_3	0.980***	0.982***	0.980***	0.974***	0.861***	0.760***	0.843***	0.779***
	(228.13)	(203.84)	(199.01)	(151.50)	(51.46)	(30.69)	(40.59)	(28.48)
TR_{ETF}	1.203***	0.745***	1.020***	0.492***	1.248***	0.780***	1.016***	0.479***
	(23.21)	(22.42)	(15.66)	(9.74)	(27.15)	(24.67)	(15.45)	(9.05)
TR_{mkt}			1.319***	1.922***			1.494***	2.126***
			(6.82)	(11.05)			(8.15)	(11.75)
VX_{t-1}^2					0.125***	0.224***	0.147***	0.214***
					(6.87)	(8.06)	(6.50)	(7.16)

Note: ( ) represents the Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

The GARCH, GJR-GARCH, and E-GARCH models correspond to equations (2), (3), and (4), respectively.

Table A4. The impacts of ETFs on stock market volatility during different periods

mean equation: $R_t = \alpha_0 + \alpha_1 R_{t-1} + \alpha_2 TR_{mkt,t} + \varepsilon_t$							
	θ_0	θ_1/φ_1	γ_1/φ_2	ω_1/φ_3	TR_{ETF}	TR_{mkt}	VX_{t-1}^2
GARCH							
2014.1.1-2023.12.31	0.6667***	0.0650***		0.5604***	0.3338***	0.6150***	-0.0435**
	(4.21)	(4.13)		(5.08)	(3.73)	(3.06)	(-2.38)
2014.1.1-2016.11.1	0.5468	0.0193		0.5758**	0.2389*	0.5741*	-0.0348
	(1.59)	(1.13)		(2.08)	(1.69)	(1.71)	(-1.10)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

2016.11.2-2020.1.29	0.1990***	0.1533***		0.7108***	0.3166***	0.0459	-0.0246*
	(2.62)	(4.43)		(18.77)	(10.34)	(0.43)	(-1.71)
2020.1.30-2022.2.23	-0.3435	0.1186***		0.6795***	0.7584***	-0.380	0.0874
	(-1.10)	(3.85)		(10.24)	(6.95)	(-1.40)	(1.52)
2022.2.24-2023.12.31	0.4277*	0.1707***		0.6797***	0.7468***	0.5343***	-0.0512
	(1.90)	(3.19)		(9.62)	(8.23)	(8.45)	(-1.25)
GJR-GARCH							
2014.1.1-2023.12.31	0.4851***	0.0583***	0.1982***	0.5675***	0.3710***	0.0864**	-0.050***
	(5.09)	(2.93)	(5.76)	(14.03)	(19.11)	(2.51)	(-2.99)
2014.1.1-2016.11.1	0.4906*	0.0595	-0.0279	0.5683***	0.1723*	0.4715**	-0.0394
	(1.81)	(1.23)	(-0.63)	(2.55)	(1.73)	(2.45)	(-1.27)
2016.11.2-2020.1.29	0.1257***	-0.0152*	0.2551***	0.6952***	0.3430***	0.0021	-0.0063
	(3.92)	(-1.47)	(5.53)	(17.91)	(11.51)	(0.02)	(-1.11)
2020.1.30-2022.2.23	1.1642**	0.0412	0.1099	0.5672***	1.9739***	1.2436	-0.0418
	(2.15)	(0.40)	(0.94)	(3.75)	(5.23)	(1.37)	(-0.60)
2022.2.24-2023.12.31	-0.1874	-0.0054	0.0959***	0.8451***	0.7182***	0.9041***	0.0477
	(-1.13)	(-0.23)	(3.37)	(28.75)	(7.02)	(3.54)	(1.47)
E-GARCH							
2014.1.1-2023.12.31	-1.286***	0.0828**	-0.152***	0.8010***	1.0051***	1.3554***	0.1980***
	(-7.53)	(2.52)	(-7.68)	(31.79)	(14.26)	(6.97)	(7.08)
2014.1.1-2016.11.1	-1.614***	0.0972	-0.113***	0.7220***	0.9079***	2.3784***	0.2488***
	(-3.86)	(1.33)	(-2.68)	(11.93)	(6.84)	(5.82)	(3.43)
2016.11.2-2020.1.29	-1.788***	-0.0381	-0.0935**	0.6711***	1.0145***	1.3267***	0.2842***
	(-4.73)	(-0.52)	(-2.22)	(12.45)	(7.81)	(4.19)	(4.51)
2020.1.30-2022.2.23	-2.061***	0.1028	-0.198***	0.7689***	1.3297***	-0.3221	0.3135***
	(-3.60)	(0.96)	(-3.54)	(13.07)	(6.76)	(-0.60)	(3.33)
2022.2.24-2023.12.31	-2.055***	-0.0155	-0.188***	0.7826***	1.2992***	1.5538**	0.3426***
	(-3.50)	(-0.14)	(-3.24)	(14.84)	(5.87)	(2.33)	(3.43)

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

Note: ( ) represents the Z statistics, *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

The GARCH, GJR-GARCH, and E-GARCH models correspond to equations (2), (3), and (4), respectively.

DOI：10.6160/SYSMR.202503/PP.0002

airiti ARTICLES IN PRESS

ABOUT THE AUTHOR

E-Ching Wu

He is an Assistant Professor in the Department of Finance at Providence University, Taiwan. He received his Ph.D. in Finance from National Central University, Taiwan. His research interests include asset management, market efficiency, and broader topics in investment. His work has been published in journals such as *Australian Economic Papers*, *Mathematics and Computers in Simulation*, *Journal of Mathematical Finance*, and *Review of Quantitative Finance and Accounting*.

Email: wuec@pu.edu.tw

DOI：10.6160/SYSMR.202503/PP.0002